UNIVERSITI TEKNOLOGI MALAYSIA

Subject : **Database(SECD2523)**

Section : **03**

Task : **Phase 3 - Database Logical Design**

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Table of Contents

1.0 Introduction

In the era of rapid e-commerce growth, the volume of parcels delivered to University Teknology Malaysia has increased exponentially. Traditional parcel management centers often rely on manual logging methods or fragmented systems to handle incoming deliveries for students and staff. These manual processes are prone to human error, resulting in misplaced items, inefficient retrieval times, and congestion at collection counters. Furthermore, the lack of real-time tracking and automated notifications often leads to delayed collections and overcrowded storage facilities.

To address these operational challenges, this project proposes the development of a "Parcel Service System" for Cengal Parcel Point (CPP). This system is a centralized, database-driven solution designed to automate the lifecycle of parcel processing from initial receipt and logging to secure storage and final collection. By integrating Smart Locker technology with a robust relational database, the system aims to streamline operations, enhance data integrity, and provide a seamless, self-service experience for the university community.

2.0 Overview of Project

This project focuses on the comprehensive design and implementation of the backend database architecture for the Parcel Service System. The development process began with Logical Database Design, where the initial conceptual model was transformed into a technical logical schema. This phase involved mapping entities to relational tables and meticulously structuring Superclass/Subclass relationships for distinct user groups, specifically separating Students from Staff and Administrators from Operational Staff. This approach allowed for the efficient management of shared attributes while enforcing specific constraints for distinct roles, thereby eliminating null value redundancies.

Following the structural design, a rigorous Normalization process was applied to ensure the highest standard of data integrity. The database schema was refined from First Normal Form (1NF) through to Boyce-Codd Normal Form (BCNF). This involved splitting composite attributes, such as names and addresses, into atomic values and resolving multi-valued attributes—like staff phone numbers and report types—by creating dedicated bridge tables. These steps were critical in removing transitive dependencies and ensuring that the database remains free of insertion, update, and deletion anomalies.

The final phase of the project centered on Physical Implementation using Structured Query Language (SQL). The normalized schema was translated into functional code through the development of Data Definition Language (DDL) scripts. This included writing precise CREATE TABLE commands for all fourteen entities, establishing Primary and Foreign Key constraints to maintain referential integrity, and defining strict data types to optimize storage. By moving from abstract design to concrete SQL implementation, this project delivers a scalable and fully functional database foundation capable of supporting the real-world logistical demands of the university.

3.0 Database Conceptual Design

3.1 Updated Business Rules

3.1.1 **Parcel Handling & Logging Rules**

- All incoming Parcel must be logged digitally into the central database by an authenticated CPP Staff account.
- A valid Parcel entry must contain a unique parcelID, trackingNumber, courierName, parcelSize, and must be linked to a verified User (recipient) via their UTMID.
- The system must automatically assign the Parcel to an available Locker unit based on ParcelSize upon logging.
- Each parcel can only be handled by one staff and assigned to one locker at a time
- If the parcel is oversized or intended for counter collection, it must not be assigned a locker (lockerID = NULL).
- Each parcel can generate multiple notifications (Arrival, Ready for Collection, Reminder, Overdue), but exactly one payment record if a fee is incurred.

3.1.2 **Customer & Notification Rules**

- Once a Parcel is assigned a Locker and its status is 'Ready for Collection', the system must automatically send an electronic Notification to the Customer's registered phone number.
- The Customer entity must be specialized into two subtypes, Student and Staff, each inheriting the common UTMID but storing subtype-specific attributes.
- A single parcel may generate multiple notifications across different statuses.
- Each notification must be linked to exactly one parcel and exactly one customer.
- Notification for **payment receipt** is sent only after a payment is successfully completed.

3.1.3 **Collection & Transaction Rules**

- To collect a parcel, the Customer must authenticate via the locker or counter using their UTMID and ParcelID and scan the QR code linked to parcelID that the system auto-generates.
- A successful Collection using the Locker interface must automatically update the corresponding Parcel record's parcelStatus to 'Collected', and the Locker Status shall change to "Available".
- If a parcel is collected after the allowed period, the system shall automatically record a penaltyFee in the Payment entity and mark the parcel as "Overdue".
- Every payment record must link directly to the corresponding Parcel, Customer, and the Staff who processed it.
- Each parcel can have separate payments. A customer can make multiple payments for one parcel.
- After payment is completed, the system sends a payment confirmation notification

3.1.4 **Maintenance & Integrity**

- Only authenticated Staff accounts can insert, update, or delete Parcel and Payment records. Customers have read-only access to their own parcel history.
- Every creation, update, or deletion of a Parcel record must be linked to the performing Staff's StaffID and a precise timestamp for full audit trail.
- If a payment is made, the Payment record must link directly to its Parcel and include the amount and paymentDate.
- Inactive or completed customer data may be archived only after all related parcels and payments are closed.
- Customers can only view their own parcel status, notifications, and payment history.

3.2 Conceptual ERD

https://app.diagrams.net/#G10H6xX2TPf60KWcuKL06RD4WMCWtEV7LI#%7B%22pageId%22%3A%226GACXYRyp226Lz3YDWqD%22%7D

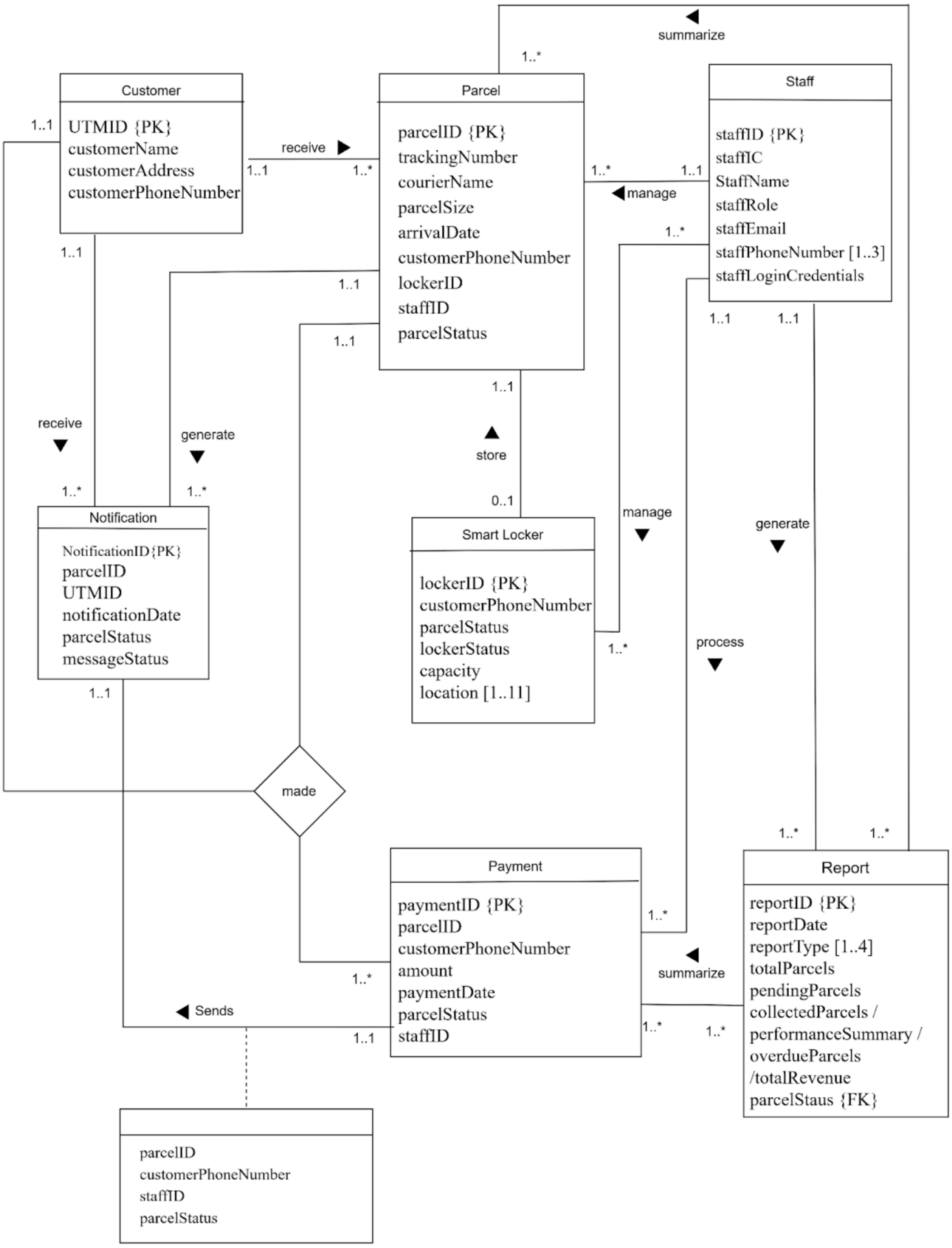


Figure 3.2.1: Conceptual ERD

4.0 Database Logical Design

4.1 Logical ERD

https://app.diagrams.net/#G1R6Iw7Uw4WbSVEglbng8N-ZJjcHrqUeWH#%7B%22pageId%22%3A%22skUbYuG2eaCn7t469jRm%22%7D

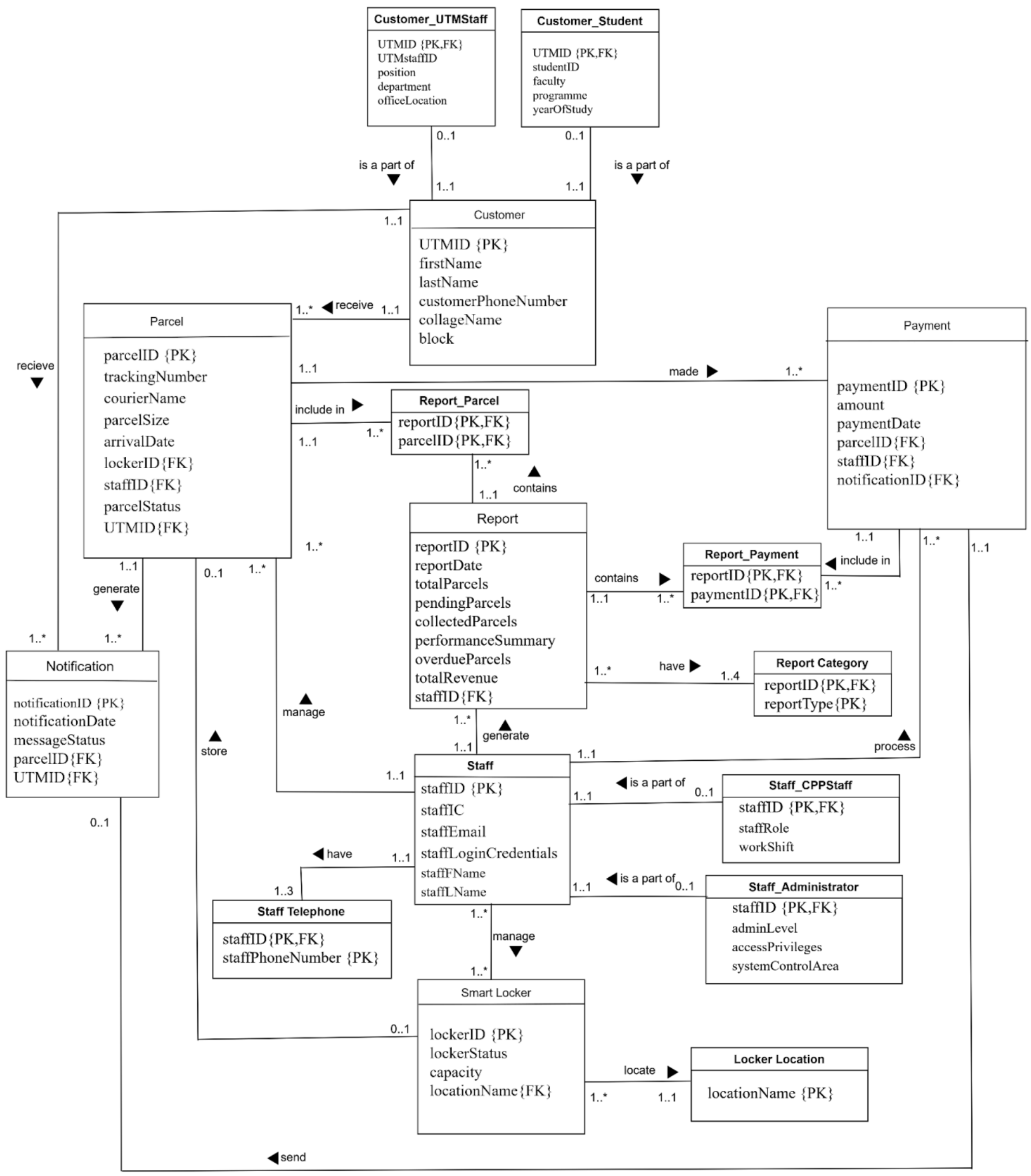


Figure 4.1.1: Logical ERD

4.2 Updated Data Dictionary

Description of Entity

Entity	Description	Occurence
Parcel	Contains information about each parcel handled by CPP, including tracking number, courier name, parcel size, arrival date, and parcel status. Each parcel is linked to one customer, assigned to one smart locker, managed by one CPP staff, and generates a payment for service or overdue.	One parcel belongs to exactly one customer. One parcel may or may not be assigned to a locker (depending on size). One parcel generates multiple notifications. One parcel can have multiple payment records.
SmartLocker	Records information for all smart lockers used for parcel storage, including locker ID, capacity, locker status, and location. Smart lockers are automatically updated when parcels are assigned or collected.	A locker can store zero or one parcel at a time. Over time, a locker can store many different parcels. It links only to the Parcel currently stored inside. A Smart Locker can be managed by one or more staff.
Locker Location	Stores the specific physical locations or zones of the smart lockers.	One Smart Locker can only be allocated at one location. There are only a maximum of 11 locations in UTM to allocate the Smart Locker.
Customer	Stores the details for all parcel recipients such as UTM ID, name, address, and contact number. A Customer can be either a Student or a UTM Staff. Customers receive parcels and are automatically notified upon parcel arrival or locker assignment.	One customer can receive one or many parcels. One customer can receive multiple notifications. One customer can make multiple payments for one parcel.
Customer_ Student	Store attribute specific to student, such as studentID, faculty, programme,and year of study. Linked to Customer via UTMID.	One Student corresponds to exactly one Customer record.
Customer_ UTMStaff	Store attribute specific to UTM staff customer, such as UTMstaffID, position,department, and office	One UTM Staff corresponds to exactly one Customer record.

	location. Linked to Customer via UTMID.	
Payment	Stores transaction details for all payments, such as payment ID, amount and payment date. Payments are linked to specific parcels and staff members who process them.	One payment is linked to exactly one parcel. One payment sends exactly one confirmation notification. One payment is processed by one staff member.
Staff	Contains details of CPP staff and administrators who manage system operations including staffID, IC, name, email, phone number and login credentials.	One staff member can manage many parcels, payments, and reports. One staff member can process many payments. One staff member can generate many reports.
Staff Telephone	Stores the work contact numbers for Management staff. The phone numbers from the main Management table to allow for multiple contacts per staff member (e.g., office, mobile).	One staff member can have multiple phone numbers (1 to 3). Each phone number record belongs to exactly one staff member.
Staff_ CPPStaff	A staff member can log parcels, assign lockers, update parcel status, and process payments. Stores attributes specific to operational staff, such as work shift.	One CPP Staff member corresponds to exactly one Management record. Linked to Management via staffID.
Staff_ Administrator	An administrator can generate reports, manage user accounts, and maintain system settings. Stores attributes specific to administrators, such as admin level and access privileges.	One Administrator corresponds to exactly one Management record. Linked to Management via staffID.
Report	Summarizes system performance and operational data such as total parcels, overdue parcels, collected parcels, and total revenue. Generated by the administrator for performance analysis and decision-making.	One report can summarize one or multiple parcel and payment records. One administrator generates one or many reports.
Report Category	Stores the categories or classifications (e.g., Performance, Revenue) applicable to a specific report. This allows a single report to cover multiple topics without repeating columns.	One Report can be classified under one or many types (1 to 4). Each report type entry is linked to exactly one generated Report.

Report_Parcel	An intersection table created to resolve the Many-to-Many relationship between Report and Parcel. It links a specific report to the exact parcel records it summarizes.	One row represents one specific parcel being included in one specific report.
Report_ Payment	An intersection table created to resolve the Many-to-Many relationship between Report and Payment. It links a specific report to the exact payment records it summarizes.	One row represents one specific payment being included in one specific report.
Notification	Records information about notifications sent to customers, including notification ID, UTM ID, parcel ID, date sent, message status, and related. Used to track and verify that customers are informed about parcel arrivals or ready-for-collection updates.	One parcel can generate multiple notifications, and each notification is linked to one parcel and one customer. Customers may receive one or many notifications. Notification may or may not be linked to a Payment.

Entity	Attributes
Parcel	parcelID {PK}, trackingNumber, courierName, parcelSize, arrivalDate, lockerID {FK}, staffID {FK}, UTMID {FK} and parcelStatus
SmartLocker	lockerID {PK}, lockerStatus, capacity, and locationName {FK}
LockerLocation	locationName{PK}
Customer (Superclass)	UTMID {PK}, firstName, lastName, collegeName,block and customerPhoneNumber
Customer_UTMStaff (Subclass)	UTMID {PK,FK}, UTMstaffID, position, department and officeLocation
Customer_Student (Subclass)	UTMID {PK,FK}, studentID, faculty, programme and yearOfStudy
Payment	paymentID {PK}, parcelID {FK}, amount, paymentDate, notificationID {FK} and staffID {FK}
Staff (Superclass)	staffID {PK}, staffIC, staffFName, staffLName, staffEmail and staffLoginCredentials
StaffTelephone	staffID {PK,FK}, staffPhoneNumber {PK}
Staff_CPPStaff (Subclass)	staffID{PK,FK}, staffRole and workShift
Staff_Administrator (Subclass)	staffID{PK,FK}, adminLevel, accessPrivileges and systemControlArea
Report	reportID {PK}, reportDate, totalParcels, pendingParcels, collectedParcels, performanceSummary, overdueParcels, totalRevenue and staffID {FK}
ReportCategory	reportType{PK} and reportID {PK,FK}
Report_Payment	paymentID(PK,FK) and reportID {PK,FK}
Report_Parcel	parcelID{PK,FK} and reportID {PK,FK}
Notification	notificationID {PK}, parcelID {FK}, UTMID {FK}, notificationDate and messageStatus

Description of Relationship

Entity 1	Multiplicity	Relationship	Entity 2	Multiplicity
Customer	1..1	receive	Notification	1..*
	1..1	receive	Parcel	1..*
Customer_Student	0..1	is a part of	Customer	1..1
Customer_ UTMStaff	0..1	is a part of	Customer	1..1
Staff	1..1	manage	Parcel	1..*
	1..*	manage	Smart Locker	1..*
	1..1	generate	Report	1..*
	1..1	process	Payment	1..*
	1..1	have	Staff Telephone	1..3
Staff_ CPPStaff	0..1	is a part of	Management	1..1
Staff_ Administrator	0..1	is a part of	Management	1..1
Parcel	1..1	generate	Notification	1..*
	1..1	made	Payment	1..*
	1..1	include in	Report_Parcel	1..*
SmartLocker	0..1	store	Parcel	0..1
	1..*	locate	Locker Location	1..1
Report	1..*	summarize	Parcel	1..*
	1..*	summarize	Payment	1..*
	1..*	have	Report Category	1..4
	1..1	contains	Report_Payment	1..*
	1..1	contains	Report_Parcel	1..*
Payment	1..1	sends	Notification	0..1
	1..1	include in	Report_Payment	1..*

Description of Attributes

Entity	Attributes	Description
Parcel	parcelID	A unique, system-generated identifier for the parcel. It is a **Primary Key** of **Parcel** entity.
	trackingNumber	The unique tracking number assigned by the external courier service.
	courierName	The name of the courier company that delivered the parcel.
	parcelSize	The size category of the parcel (e.g., 'Small', 'Large'), used for locker assignment. Small parcel will go to smart locker, big parcel that over locker size will go to CPP counter.
	arrivalDate	The date the parcel was officially received and logged into the system by the staff.
	lockerID	The identifier of the specific **Smart Locker** unit the parcel is assigned to for collection. It is a **Foreign Key** references to **Locker(lockerID)**.
	staffID	The identifier of the **Management** member who performed the initial logging of the parcel. It is a **Foreign Key** references to **Management(staffID)**.
	parcelStatus	The current status of the parcel (e.g., 'Arrived', 'Ready for Collection', 'Collected', 'Overdue').
	UTMID	Identify the parcel owner. It is a **Foreign Key** reference to **Customer(UTMID).**
SmartLocker	lockerID	A unique identifier for each physical **Smart Locker** unit in the system. It is a **Primary Key** for **Smart Locker** entity.
	lockerStatus	The status of the locker. If there is parcel inside the locker, status will become "Occupied", if there is no parcel inside the locker, status will become "Available".
	capacity	The maximum number of smart locker in UTM.
	locationName	It is a **Foreign Key** reference to **Locker Location(locationName).** Ensures the locker is

		assigned to one of the 11 valid spots.
Locker Location	locationName	The unique name of the location (e.g., 'KTHO', 'Arcade'). It is a **Primary Key** for **Locker Location** entity. This table ensures only valid locations are assigned to lockers
Customer (Superclass)	UTMID	The unique UTM ID of the individual serves as the main identifier for logins. It is a **Primary Key** for **Customer** entity
	firstName	The customer first name
	lastName	The customer last name
	collegeName	The name of the residential college or building.
	block	The specific block number of the address.
	customerPhoneNumber	The primary contact number of the recipient, used for manual lookups and automated notifications
Customer_ Student	UTMID	It is a **Foreign Key** reference to **Customer(UTMID)**. It is also a **Primary Key** to link the student to their user record.
	studentID	The unique matric card number.
	faculty	The faculty the student belongs to.
	programme	The academic programme code that the student enrolls in.
	yearOfStudy	The student’s current year of study.
Customer_ UTMStaff	UTMID	It is a **Foreign Key** reference to **Customer(UTMID)**. It is also a **Primary Key** to link the staff customer to their user record.
	UTMstaffID	The unique staff ID card number.
	position	The job title of the staff member that works in UTM.
	department	The department the staff belongs to.
	officeLocation	The physical location of the staff's office.

Payment	paymentID	A unique, system-generated identifier for the payment transaction. It is a **Primary Key** for **Payment** entity.
	parcelID	The identifier of the **Parcel** for which the penalty or fee was incurred, linking the payment directly to the parcel. It is a **Foreign Key** references to **Parcel(ParcelID)**
	amount	The total monetary value of the payment transaction
	paymentDate	The date and time the payment was officially recorded
	staffID	The identifier of the **Management** member who processed or verified the payment. It is a **Foreign Key** references to **Management(staffID)**
	notificationID	The identifier of the **Notification** when the customer made a payment. It is a **Foreign Key** reference to **Notification(notificationID).** Links to the mandatory confirmation receipt
Staff (Superclass)	staffID	A unique identifier for the management member. It is a **Primary Key** for **Management** entity
	staffIC	The staff member's National Identification Card (IC) number
	staffFname	The staff's first name.
	staffLName	The staff's last name.
	staffEmail	The official UTM email address for system communication and verification
	staffLoginCredentials	A hashed and secured representation of the staff member's password or login key
Staff Telephone	staffID	It is a part of **Composite Primary Keys**. It is a **Foreign Key** reference to **Management(staffID),** which can identify the phone number.
	staffPhoneNumber	The second component of the **Composite Primary Key**. It stores the actual contact

		number. By making this part of the PK, the database allows one **staffID** to exist multiple times (one for each distinct phone number).
Staff_ CPPStaff (Subclass)	staffID	The identifier for the operational staff. It acts as both the **Primary Key** for this table and a **Foreign Key** to the **Management(staffID)** superclass.
	staffRole	The role assigned to a staff member.
	workShift	The assigned working shift pattern (e.g., '08:00-16:00', '16:00-00:00') for the staff member.
Staff_ Administrator (Subclass)	staffID	The identifier for the administrator. It acts as both the **Primary Key** for this table and a **Foreign Key** to the **Management(staffID)** superclass.
	adminLevel	The hierarchical clearance level (e.g., 'SuperAdmin', 'Supervisor') determining the scope of authority.
	accessPrivileges	Detailed system permissions (e.g., 'Read-Only', 'Full-Edit') granted to this administrator.
Report	reportID	Unique identifier for each generated report. It is a **Primary Key** for **Report** entity
	reportDate	The date the report was generated or the period it covers
	totalParcels	The total number of parcels logged during the reporting period
	pendingParcels	The count of parcels that were available for collection but not yet collected
	collectedParcels	The total number of parcels collected during the reporting period
	performanceSummary	A brief, summarized text field describing the overall performance metrics
	overdueParcels	The count of parcels that have exceeded the allowed collection time

	totalRevenue	The total revenue collected from penalty fees during the reporting period
	staffID	It is a **Foreign Key** references to **Management(staffID).** This records exactly which Administrator authorized and generated the report.
Report Category	reportID	The first part of the **Composite Primary Key**. It serves as a **Foreign Key** linking back to the **Report(reportID)** table, identifying which report these categories belong to.
	reportType	The second part of the **Composite Primary Key**. It stores the specific classification. The composite key structure allows a single **reportID** to be tagged with multiple distinct categories while preventing duplicate tags for the same report.
Report_ Payment	reportID	The first part of the **Composite Primary Key**. It is a **Foreign Key** referencing **Report(reportID)**. It identifies which report the payment belongs to.
	paymentID	The second part of the **Composite Primary Key**. It is a **Foreign Key** referencing **Payment(paymentID)**. It identifies the specific payment record included in the report.
Report_Parcel	reportID	The first part of the **Composite Primary Key**. It is a **Foreign Key** referencing **Report(reportID)**. It identifies which report the parcel belongs to.
	parcelID	The second part of the **Composite Primary Key**. It is a **Foreign Key** referencing **Parcel(parcelID)**. It identifies the specific parcel record included in the report.
Notification	NotificationID	A unique, system-generated identifier for the notification instance. It is a **Primary Key** for Notification entity
	parcelID	The identifier of the **Parcel** that the notification is about. It is a **Foreign Key** references to **Parcel(parcelID)**
	UTMID	The identifier of the **Customer** (recipient) the

		notification was sent to. It is a **Foreign Key** references to **Customer(UTMID)**
	notificationDate	The notification date that sent to customer
	messageStatus	The status of the message delivery to customer

Entity	Attribute	Description	Data Type	Null	Multi-Valued
Parcel	parcelID	Uniquely identifies each parcel.(PK)	VARCHAR(10)	No	No
	trackingNumber	Unique tracking number assigned by courier.	VARCHAR(20)	No	No
	courierName	Name of the courier company.	VARCHAR(15)	No	No
	parcelSize	Size of the parcel (Small/Large).	VARCHAR(10)	No	No
	arrivalDate	Date that parcel was received by CPP.	DATE	No	No
	lockerID	Locker assigned to the parcel.(FK)	VARCHAR(10)	Yes	No
	staffID	Staff ID who logged the parcel.(FK)	VARCHAR(10)	No	No
	parcelStatus	Current parcel status (Arrived/Collected /Overdue).	VARCHAR(20)	No	No
	UTMID	Receiver ID. (FK)	VARCHAR(10)	No	No
Smart Locker	lockerID	Unique identifier for each locker. (PK)	VARCHAR(10)	No	No
	lockerStatus	Locker availability (Available/ Occupied).	VARCHAR(15)	No	No
	capacity	Maximum number of lockers.	INT	No	No
	locationName	Physical location of locker. (FK)	VARCHAR(50)	No	No
Locker Location	locationName	Valid location name. (PK)	VARCHAR(50)	No	No

Customer	UTMID	Unique UTM identifier for the customer.(PK)	VARCHAR(10)	No	No
	firstName	User's first name.	VARCHAR(50)	No	No
	lastName	User's last name.	VARCHAR(50)	No	No
	collegeName	Residential college name	VARCHAR(50)	No	No
	block	Block number of address.	VARCHAR(20)	No	No
	customerPhoneNumber	Primary contact number of the customer.	VARCHAR(15)	No	No
Customer_ Student	UTMID	References User ID. (PK, FK)	VARCHAR(10)	No	No
	studentID	Unique matric card ID.	VARCHAR(10)	No	No
	faculty	Faculty name.	VARCHAR(50)	No	No
	programme	Programme code.	VARCHAR(10)	No	No
	yearOfStudy	Current year of study.	INT	No	No
Customer_ UTMStaff	UTMID	References User ID. (PK, FK)	VARCHAR(10)	No	No
	UTMstaffID	Unique staff card ID.	VARCHAR(10)	No	No
	position	Job title.	VARCHAR(50)	No	No
	department	Department name.	VARCHAR(50)	No	No
	officeLocation	Office room or zone.	VARCHAR(50)	No	No
Payment	paymentID	Unique payment transaction ID. (PK)	VARCHAR(10)	No	No

	parcelID	Parcel linked to the payment.(FK)	VARCHAR(10)	No	No
	amount	Total payment amount.	DECIMAL(10,2)	No	No
	paymentDate	Date payment was made.	DATE	No	No
	staffID	Staff who processed payment.(FK)	VARCHAR(10)	No	No
	notificationID	Confirmation notification ID. (FK)	VARCHAR(10)	No	No
Staff	staffID	Unique identifier for CPP staff.(PK)	VARCHAR(10)	No	No
	staffIC	Staff IC number.	VARCHAR(12)	No	No
	staffFName	Staff first name.	VARCHAR(50)	No	No
	staffLName	Staff last name.	VARCHAR(50)	No	No
	staffEmail	Official email for login.	VARCHAR(50)	No	No
	staffLoginCredentials	Encrypted login credentials.	VARCHAR(50)	No	No
Staff Telephone	staffID	References Management. (PK, FK)	VARCHAR(10)	No	No
	staffPhoneNumber	Contact number. (PK)	VARCHAR(15)	No	No
Staff_ CPPStaff	staffID	References Management. (PK, FK)	VARCHAR(10)	No	No
	staffRole	Assigned role	VARCHAR(50)	No	No
	workShift	Assigned shift time.	VARCHAR(20)	No	No
Staff_ Administrator	staffID	References Management. (PK, FK)	VARCHAR(10)	No	No
	adminLevel	Admin clearance level.	VARCHAR(50)	No	No
	accessPrivileges	System access rights.	VARCHAR(20)	No	No

Report	reportID	Unique report identifier.(PK)	VARCHAR(10)	No	No
	reportDate	Date report was generated.	DATE	No	No
	totalParcels	Total parcels recorded in the report.	INT	No	No
	pendingParcels	Count of pending parcels.	INT	Yes	No
	collectedParcels	Count of collected parcels.	INT	Yes	No
	performanceSummary	Summary of CPP performance.	VARCHAR(200)	No	No
	overdueParcels	Count of overdue parcels.	INT	Yes	No
	totalRevenue	Total payment revenue collected.	DECIMAL(10,2)	Yes	No
	staffID	Generating Admin ID. (FK)	VARCHAR(10)	No	No
Report Category	reportID	References Report. (PK, FK)	VARCHAR(10)	No	No
	reportType	Category (Revenue/Performance). (PK)	VARCHAR(20)	No	No
Report_ Payment	reportID	References Report. (PK, FK)	VARCHAR(10)	No	No
	paymentID	References Payment (PK,FK)	VARCHAR(10)	No	No
Report_Parcel	reportID	References Report. (PK, FK)	VARCHAR(10)	No	No
	parcelID	References Parcel. (PK,FK)	VARCHAR(10)	No	No
Notification	notificationID	Unique notification record ID.(PK)	VARCHAR(10)	No	No
	parcelID	Parcel associated with notification.(FK)	VARCHAR(10)	No	No
	UTMID	Recipient of the	VARCHAR(10)	No	No

		notification.(FK)			
	notificationDate	Date notification was sent.	DATE	No	No
	messageStatus	Delivery status of message.	VARCHAR(20)	No	No

4.3 Normalization

1. Parcel (parcelID, trackingNumber, courierName, parcelSize, arrivalDate, lockerID, staffID, parcelStatus, UTMID)
   fd1: parcelID → trackingNumber, courierName, parcelSize, arrivalDate, lockerID, staffID, parcelStatus, UTMID
   1NF&2NF&3NF&BNCF:
   Parcel (<u>parcelID</u>, trackingNumber, courierName, parcelSize, arrivalDate, lockerID, staffID, parcelStatus, UTMID)

2. SmartLocker (lockerID, lockerStatus, capacity, locationName)
   fd1: lockerID → lockerStatus, capacity, locationName
   1NF&2NF&3NF&BNCF:
   SmartLocker (<u>lockerID</u>, lockerStatus, capacity, locationName)

3. LockerLocation (locationName)
   fd1: locationName → locationName (trivial)
   1NF&2NF&3NF&BNCF:
   LockerLocation (<u>locationName</u>)

4. Customer (UTMID, firstName, lastName, collegeName, block, customerPhoneNumber)
   fd1: UTMID → firstName, lastName, collegeName, block, customerPhoneNumber
   1NF&2NF&3NF&BNCF:
   Customer (<u>UTMID</u>, firstName, lastName, collegeName, block, customerPhoneNumber)

5. Customer_Student (UTMID, studentID, faculty, programme, yearOfStudy)
   fd1: UTMID → studentID, faculty, programme, yearOfStudy
   1NF&2NF&3NF&BNCF:
   Customer_Student (<u>UTMID</u>, studentID, faculty, programme, yearOfStudy)

6. Customer_UTMStaff (UTMID,UTMstaffID, position, department, officeLocation)
   fd1: UTMID → UTMstaffID, position, department, officeLocation

1NF&2NF&3NF&BNCF:

Customer_UTMStaff (UTMID, UTMstaffID, position, department, officeLocation)

7. Payment (paymentID, parcelID, amount, paymentDate, staffID, notificationID)

   fd1: paymentID → parcelID, amount, paymentDate, staffID, notificationID

   1NF&2NF&3NF&BNCF:

   Payment (paymentID, parcelID, amount, paymentDate, staffID, notificationID)

8. Staff (staffID, staffIC, staffFName, staffLName, staffEmail, staffLoginCredentials)

   fd1: staffID → staffIC, staffFName, staffLName, staffEmail, staffLoginCredentials

   1NF&2NF&3NF&BNCF:

   Staff (staffID, staffIC, staffFName, staffLName, staffEmail, staffLoginCredentials)

9. StaffTelephone (staffID, staffPhoneNumber)

   fd1: {staffID, staffPhoneNumber} → {staffID, staffPhoneNumber} (trivial)

   1NF&2NF&3NF&BNCF:

   StaffTelephone (staffID, staffPhoneNumber)

10. Staff_CPPStaff (staffID, staffRole, workShift)

    fd1: staffID → staffRole, workShift

    1NF&2NF&3NF&BNCF:

    Staff_CPPStaff (staffID, staffRole, workShift)

11. Staff_Administrator (staffID, adminLevel, accessPrivileges)

    fd1: staffID → adminLevel, accessPrivileges

    1NF&2NF&3NF&BNCF:

    Staff_Administrator (staffID, adminLevel, accessPrivileges)

12. Report (reportID, reportDate, totalParcels, pendingParcels, collectedParcels, performanceSummary, overdueParcels, totalRevenue, staffID)

    fd1: reportID → reportDate, totalParcels, pendingParcels, collectedParcels, performanceSummary, overdueParcels, totalRevenue, staffID

1NF&2NF&3NF&BNCF:

Report (reportID, reportDate, totalParcels, pendingParcels, collectedParcels, performanceSummary, overdueParcels, totalRevenue, staffID)

13. ReportCategory (reportID, reportType)

    fd1: {reportID, reportType} → {reportID, reportType} (trivial)

    1NF&2NF&3NF&BNCF:

    ReportCategory (reportID, reportType)

14. Report_Payment (reportID, paymentID)

    fd1: {reportID, paymentID} → {reportID, paymentID} (trivial)

    1NF&2NF&3NF&BNCF:

    Report_Payment (reportID, paymentID)

15. Report_Parcel (reportID, parcelID)

    fd1: {reportID, parcelID} → {reportID, parcelID} (trivial)

    1NF&2NF&3NF&BNCF:

    Report_Parcel (reportID, parcelID)

16. Notification (notificationID, parcelID, UTMID, notificationDate, messageStatus)

    fd1: notificationID → parcelID, UTMID, notificationDate, messageStatus

    1NF&2NF&3NF&BNCF:

    Notification (notificationID, parcelID, UTMID, notificationDate, messageStatus)

*Remark: Underline word is primary key.

*Remark: trivial - Since there are no non-key attributes, there are no dependencies to check. The key simply determines itself.

5.0 Relational DB Schemas (after normalization)

The relational database schema for "Parcel Service System" for Cengal Parcel Point (CPP) is consist of:

Parcel	(parcelID, trackingNumber, courierName, parcelSize, arrivalDate, lockerID, staffID, parcelStatus, UTMID)
SmartLocker	(lockerID, lockerStatus, capacity, locationName)
LockerLocation	(locationName)
Customer	(UTMID, firstName, lastName, collegeName, block, customerPhoneNumber)
Customer_Student	(UTMID, studentID, faculty, programme, yearOfStudy)
Customer_UTMStaff	(UTMID, UTMstaffID, position, department, officeLocation)
Payment	(paymentID, parcelID, amount, paymentDate, staffID, notificationID)
Staff	(staffID, staffIC, staffFName, staffLName, staffEmail, staffLoginCredentials)
StaffTelephone	(staffID, staffPhoneNumber)
Staff_CPPStaff	(staffID, staffRole, workShift)
Staff_Administrator	(staffID, adminLevel, accessPrivileges)
Report	(reportID, reportDate, totalParcels, pendingParcels, collectedParcels, performanceSummary, overdueParcels, totalRevenue, staffID)
ReportCategory	(reportID, reportType)

Report_Payment	(reportID, paymentID)
Report_Parcel	(reportID, parcelID)
Notification	(notificationID, parcelID, UTMID, notificationDate, messageStatus)

*Remark: Underline word is primary key.

Parcel

parcelID	trackingNumber	courierName	parcelSize	arrivalDate	lockerID
staffID	parcelStatus	UTMID			

SmartLocker

lockerID	lockerStatus	capacity	locationName

LockerLocation

locationName

Customer

UTMID	firstName	lastName	collegeName	block	customerPhoneNumber

Customer_Student

UTMID	studentID	faculty	programme	yearOfStudy

Customer_UTMStaff

UTMID	UTMstaffID	position	department	officeLocation

Payment

paymentID	parcelID	amount	paymentDate	staffID	notificationID

Staff

staffID	staffIC	staffFName	staffLName	staffEmail	staffLoginCredentials

StaffTelephone

staffID	staffPhoneNumber

Staff_CPPStaff

staffID	staffRole	workShift

Staff_Administrator

staffID	adminLevel	accessPrivileges

Report

reportID	reportDate	totalParcels	pendingParcels	collectedParcels
performanceSummary	overdueParcels	totalRevenue	staffID	

ReportCategory

reportID	reportType

Report_Payment

reportID	paymentID

Report_Parcel

reportID	parcelID

Notification

notificationID	parcelID	UTMID	notificationDate	messageStatus

6.0 SQL Statement (DDL & DML)

6.1 DDL

```
DROP DATABASE IF EXISTS cpp_database;

/* Create Database */
CREATE DATABASE cpp_database;

/* Use Database */
USE cpp_database;

/* Create table for each entity */
CREATE TABLE Customer (
	UTMID VARCHAR(10) PRIMARY KEY,
	firstName VARCHAR(50) NOT NULL,
	lastName VARCHAR(50) NOT NULL,
	collegeName VARCHAR(50) NOT NULL,
	block VARCHAR(20) NOT NULL,
	customerPhoneNumber VARCHAR(15) NOT NULL UNIQUE
);

CREATE TABLE Staff (
	staffID VARCHAR(10) PRIMARY KEY,
	staffIC VARCHAR(12) NOT NULL,
	staffFName VARCHAR(20) NOT NULL,
	staffLName VARCHAR(20) NOT NULL,
	staffEmail VARCHAR(50) NOT NULL,
	staffLoginCredentials VARCHAR(50) NOT NULL
);

CREATE TABLE StaffTelephone (
```

```sql
    staffID VARCHAR(10),
    staffPhoneNumber VARCHAR(15),
    PRIMARY KEY (staffID,staffPhoneNumber),
    CONSTRAINT telephone_fk_stafflId FOREIGN KEY (staffID) REFERENCES
    Staff(staffID)
);

CREATE TABLE Staff_CPPStaff (
    staffID VARCHAR(10) PRIMARY KEY,
    staffRole VARCHAR(20) NOT NULL,
    workShift VARCHAR(20) NOT NULL,
    CONSTRAINT cppstaff_fk_stafflId FOREIGN KEY (staffID) REFERENCES
    Staff(staffID)
);

CREATE TABLE Staff_Administrator (
    staffID VARCHAR(10) PRIMARY KEY,
    adminLevel VARCHAR(50) NOT NULL,
    accessPrivileges VARCHAR(20) NOT NULL,
    CONSTRAINT admin_fk_stafflId FOREIGN KEY (staffID) REFERENCES
    Staff(staffID)
);

CREATE TABLE Customer_Student (
    UTMID VARCHAR(10) PRIMARY KEY,
    StudentID VARCHAR(10),
    faculty VARCHAR(20) NOT NULL,
    programme VARCHAR(30) NOT NULL,
    yearOfStudy INT NOT NULL,
    CONSTRAINT custstudent_fk_utmId FOREIGN KEY (UTMID) REFERENCES
    Customer(UTMID)
```

```
);

CREATE TABLE Customer_UTMStaff (
    UTMID VARCHAR(10) PRIMARY KEY,
    UTMStaffID VARCHAR(10),
    position VARCHAR(20) NOT NULL,
    department VARCHAR(20) NOT NULL,
    officeLocation VARCHAR(30) NOT NULL,
    CONSTRAINT custstaff_fk_utmId FOREIGN KEY (UTMID) REFERENCES
    Customer(UTMID)
);

CREATE TABLE LockerLocation (
    locationName VARCHAR(50) PRIMARY KEY
);

CREATE TABLE SmartLocker (
    lockerID VARCHAR(10) PRIMARY KEY,
    lockerStatus VARCHAR(15) NOT NULL,
    capacity INT NOT NULL,
    locationName VARCHAR(50),
    CONSTRAINT locker_fk_locationname FOREIGN KEY (locationName)
    REFERENCES LockerLocation(locationName)
);

CREATE TABLE Parcel (
    parcelID VARCHAR(10) PRIMARY KEY,
    trackingNumber VARCHAR(20) NOT NULL,
    courierName VARCHAR(15) NOT NULL,
    parcelSize VARCHAR(10) NOT NULL,
    arrivalDate DATE NOT NULL,
```

```
    lockerID VARCHAR(10),
    staffID VARCHAR(10) NOT NULL,
    parcelStatus VARCHAR(20) NOT NULL,
    UTMID VARCHAR(10) NOT NULL,
    CONSTRAINT parcel_fk_utmId FOREIGN KEY (UTMID) REFERENCES
    Customer(UTMID),
    CONSTRAINT parcel_fk_lockerId FOREIGN KEY (lockerID) REFERENCES
    SmartLocker(lockerID),
    CONSTRAINT parcel_fk_staffId FOREIGN KEY (staffID) REFERENCES
    Staff(staffID)
);

CREATE TABLE Notification (
    notificationID VARCHAR(10) PRIMARY KEY,
    parcelID VARCHAR(10) NOT NULL,
    UTMID VARCHAR(10) NOT NULL,
    notificationDate DATE NOT NULL,
    messageStatus VARCHAR(20) NOT NULL,
    CONSTRAINT notification_fk_parcelId FOREIGN KEY (parcelID) REFERENCES
    Parcel(parcelID),
    CONSTRAINT notification_fk_utmlId FOREIGN KEY (UTMID) REFERENCES
    Customer(UTMID)
);

CREATE TABLE Payment (
    paymentID VARCHAR(10) PRIMARY KEY,
    parcelID VARCHAR(10) NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    paymentDate DATE NOT NULL,
    staffID VARCHAR(10) NOT NULL,
    NotificationID VARCHAR(10),
```

```
    CONSTRAINT payment_fk_parcelId FOREIGN KEY (parcelID) REFERENCES
    Parcel(parcelID),
    CONSTRAINT payment_fk_notiId FOREIGN KEY (NotificationID) REFERENCES
    Notification(NotificationID),
    CONSTRAINT payment_fk_stafflId FOREIGN KEY (staffID) REFERENCES
    Staff(staffID)
);

CREATE TABLE Report (
    reportID VARCHAR(10) PRIMARY KEY,
    reportDate DATE NOT NULL,
    totalParcels INT NOT NULL,
    pendingParcels INT,
    collectedParcels INT,
    overdueParcels INT,
    totalRevenue DECIMAL(10,2),
    performanceSummary VARCHAR(200) NOT NULL,
    staffID VARCHAR(10) NOT NULL,
    CONSTRAINT report_fk_stafflId FOREIGN KEY (staffID) REFERENCES
    Staff(staffID)
);

CREATE TABLE ReportCategory (
    reportID VARCHAR(10),
    reportType VARCHAR(20),
    PRIMARY KEY (reportID, reportType),
    CONSTRAINT reportcat_fk_reportId FOREIGN KEY (reportID) REFERENCES
    Report(ReportID)
);

CREATE TABLE Report_Payment (
```

```
    reportID VARCHAR(10),
    paymentID VARCHAR(10),
    PRIMARY KEY (reportID, paymentID),
    CONSTRAINT reportpay_fk_reportId FOREIGN KEY (reportID) REFERENCES
    Report(ReportID),
    CONSTRAINT reportpay_fk_paymentId FOREIGN KEY (paymentID) REFERENCES
    Payment(PaymentID)
);

CREATE TABLE Report_Parcel (
    reportID VARCHAR(10),
    parcelID VARCHAR(10),
    PRIMARY KEY (reportID, parcelID),
    CONSTRAINT reportpar_fk_reportId FOREIGN KEY (reportID) REFERENCES
    Report(ReportID),
    CONSTRAINT reportpar_fk_parcelId FOREIGN KEY (parcelID) REFERENCES
    Parcel(ParcelID)
);

/* Alter table */
ALTER TABLE SmartLocker
ADD CONSTRAINT chk_locker_status
CHECK (lockerStatus IN ('Available', 'Occupied'));

ALTER TABLE Parcel
ADD CONSTRAINT chk_parcel_status
CHECK (parcelStatus IN ('Arrived','Ready','Collected','Overdue'));

ALTER TABLE Parcel
MODIFY courierName VARCHAR(30);
```

```
ALTER TABLE Parcel
ADD CONSTRAINT uq_tracking_number UNIQUE (trackingNumber);
```

6.2 DML

TO AVOID FK ERROR, SHOULD INSERT IN THIS ORDER:

```
--LockerLocation
INSERT INTO LockerLocation (locationName) VALUES ('Kolej Tun Fatimah');
INSERT INTO LockerLocation (locationName) VALUES ('Kolej Tun Hussein Onn');
INSERT INTO LockerLocation (locationName) VALUES ('Kolej Tun Dr. Ismail');
INSERT INTO LockerLocation (locationName) VALUES ('Kolej Perdana');
INSERT INTO LockerLocation (locationName) VALUES ('Kolej  Rahman Putra');
INSERT INTO LockerLocation (locationName) VALUES ('Kolej Tuanku Canselor');
INSERT INTO LockerLocation (locationName) VALUES ('Kolej  Dato Onn Jaafar');
INSERT INTO LockerLocation (locationName) VALUES ('Kolej 9');
INSERT INTO LockerLocation (locationName) VALUES ('Kolej 10');
```

```
--SmartLocker
INSERT INTO
SmartLocker (lockerID, lockerStatus, capacity, locationName)
VALUES
('L001', 'Available', 100, 'Kolej Tuanku Canselor'),
('L002', 'Available', 80, 'Kolej Tun Hussein Onn'),
('L003','Occupied',60,'Kolej Tun Fatimah'),
('L004','Available',50,'Kolej Perdana');
--View SmartLocker Status
SELECT *
FROM SmartLocker;
```

	lockerID	lockerStatus	capacity	locationName
▶	L001	Available	100	Kolej Tuanku Canselor
	L002	Available	80	Kolej Tun Hussein Onn
	L003	Occupied	60	Kolej Tun Fatimah
	L004	Available	50	Kolej Perdana
*	NULL	NULL	NULL	NULL

```
--Customer
INSERT INTO
Customer (UTMID, firstName, lastName, collegeName, block, customerPhoneNumber)
VALUES
('UTM001', 'Ali', 'Hassan', 'Kolej Tuanku Canselor', 'Block S06', '0123456789'),
('UTM002', 'Abu', 'Bakar', 'Kolej Tun Hussein Onn', 'Block L07', '0124598624'),
('UTM003', 'Ling', 'Mei', 'Kolej Tun Fatimah', 'Block H06', '0104845719'),
('UTM004','Aisyah','Rahman','Kolej Perdana','Block D01','0192233445'),
('UTM005','Daniel','Lim','Kolej 9','Block A1','0178899001');



--Customer_Student
INSERT INTO
Customer_Student (UTMID, studentID, faculty, programme, yearOfStudy)
VALUES
('UTM001', 'A22CS001', 'FKE', 'SKEE', 2),
('UTM002', 'A22CS002', 'FC', 'SECR', 3),
('UTM004','A23CS010','FC','SECJ',1);

-- Customer_UTMStaff
INSERT INTO
Customer_UTMStaff (UTMID, UTMstaffID, position, department, officeLocation)
VALUES
('UTM003', 'UTM8899', 'Lecturer', 'FKE', 'C16-205'),
('UTM005', 'UTM8896', 'Lecturer', 'FC', 'N28-205');
--View Customer_UTMStaff
SELECT *
FROM Customer_UTMStaff;
```

UTMID	UTMStaffID	position	department	officeLocation
UTM003	UTM8899	Lecturer	FKE	C16-205
UTM005	UTM8896	Lecturer	FC	N28-205
NULL	NULL	NULL	NULL	NULL

```sql
--Staff
INSERT INTO
Staff (staffID, staffIC, staffFName, staffLName, staffEmail, staffLoginCredentials)
VALUES
('STF001', '950101015678', 'Siti','Aminah', 'siti@utm.my', 'hashed123'),
('STF002', '920505086543', 'Ahmad', 'Ali', 'ahmad@utm.my', 'hashed456'),
('STF003','880707078888','Farah','Zainal', 'farah@utm.my','hashed789');


--StaffTelephone
INSERT INTO
StaffTelephone (staffID, staffPhoneNumber)
VALUES
('STF001', '0134567890'),
('STF002', '0119876543'),
('STF003','0123344556');


--Staff_CPPStaff
INSERT INTO
Staff_CPPStaff (staffID, staffRole, workShift)
VALUES
('STF001', 'CPP Staff', 'Morning'),
('STF003','CPP Staff','Evening');


--StaffAdministrator
INSERT INTO
Staff_Administrator (staffID, adminLevel, accessPrivileges)
VALUES ('STF002', 'Level 1', 'Full Access');


--Parcel
INSERT INTO
Parcel (parcelID, trackingNumber, courierName, parcelSize, arrivalDate, lockerID, staffID,
parcelStatus, UTMID)
VALUES
```

```sql
('P001', 'TRK123456', 'DHL', 'Small', DATE '2025-06-01', 'L001', 'STF001', 'Arrived',
'UTM001'),
('P002', 'TRK654321', 'J&T', 'Large', DATE '2025-06-02', 'L002', 'STF001', 'Collected',
'UTM002'),
('P003', 'TRK888777', 'Shopee', 'Medium', DATE '2025-06-03', 'L003', 'STF003', 'Arrived',
'UTM004'),
('P004', 'TRK999666', 'PosLaju', 'Small', DATE  '2025-06-04', 'L004' ,'STF003', 'Overdue',
'UTM005');
--View Parcel Status
SELECT *
FROM Parcel;
```

parcelID	trackingNumber	courierName	parcelSize	arrivalDate	lockerID	staffID	parcelStatus	UTMID
P001	TRK123456	DHL	Small	2025-06-01	L001	STF001	Arrived	UTM001
P002	TRK654321	J&T	Large	2025-06-02	L002	STF001	Collected	UTM002
P003	TRK888777	Shopee	Medium	2025-06-03	L003	STF003	Arrived	UTM004
P004	TRK999666	PosLaju	Small	2025-06-04	L004	STF003	Overdue	UTM005
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```sql
--Notification
INSERT INTO
Notification (notificationID, parcelID, UTMID, notificationDate, messageStatus)
VALUES
('N001', 'P001', 'UTM001', DATE '2025-06-01', 'Sent'),
('N002', 'P002', 'UTM002', DATE '2025-06-02', 'Sent'),
('N003', 'P003', 'UTM004', DATE '2025-06-03', 'Sent'),
('N004', 'P004', 'UTM005', DATE '2025-06-05', 'Failed');
```

```sql
--Payment
INSERT INTO Payment
(paymentID, parcelID, amount, paymentDate, staffID, notificationID)
VALUES
('PAY001', 'P002', 5.00, '2025-06-02', 'STF001', 'N002'),
('PAY002','P004',3.00,DATE '2025-06-05','STF003','N004');
```

```sql
--Report
INSERT INTO
Report (reportID, reportDate, totalParcels, pendingParcels, collectedParcels,
performanceSummary, overdueParcels, totalRevenue, staffID)
VALUES
('R001', DATE '2025-06-30', 120, 10, 100, 'Overall parcel collection is efficient', 10, 250.00,
'STF002'),
('R002', DATE '2025-06-30', 150, 20, 120,  'Parcel processing is efficient with minor delays', 10,
320.00, 'STF002');


--ReportCategory
INSERT INTO
ReportCategory (reportID, reportType)
VALUES
('R001', 'Performance'),
('R001', 'Revenue');


 --Report_Payment
INSERT INTO
Report_Payment (reportID, paymentID)
VALUES
('R001', 'PAY001');


--Report_Parcel
INSERT INTO
Report_Parcel (reportID, parcelID)
VALUES
('R001', 'P001'),
('R001', 'P002');


/* Update parcel status */
UPDATE Parcel
SET parcelStatus = 'Collected'
WHERE parcelID = 'P001';
```

/* Show updated Parcel table */
SELECT * FROM Parcel;

parcelID	trackingNumber	courierName	parcelSize	arrivalDate	lockerID	staffID	parcelStatus	UTMID
P001	TRK123456	DHL	Small	2025-06-01	L001	STF001	Collected	UTM001
P002	TRK654321	J&T	Large	2025-06-02	L002	STF001	Collected	UTM002
P003	TRK888777	Shopee	Medium	2025-06-03	L003	STF003	Arrived	UTM004
P004	TRK999666	PosLaju	Small	2025-06-04	L004	STF003	Overdue	UTM005
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

/* Update locker status */
UPDATE SmartLocker
SET lockerStatus = 'Occupied'
WHERE lockerID = 'L001';

/* Show updated SmartLocker table */
SELECT * FROM SmartLocker;

lockerID	lockerStatus	capacity	locationName
L001	Occupied	100	Kolej Tuanku Canselor
L002	Available	80	Kolej Tun Hussein Onn
L003	Occupied	60	Kolej Tun Fatimah
L004	Available	50	Kolej Perdana
NULL	NULL	NULL	NULL

/* Delete customer dependent records */
DELETE FROM Notification WHERE UTMID = 'UTM003';
DELETE FROM Customer_UTMStaff WHERE UTMID = 'UTM003';
DELETE FROM Customer WHERE UTMID = 'UTM003';

/* Show remaining customers */
SELECT * FROM Customer;

UTMID	firstName	lastName	collegeName	block	customerPhoneNumber
UTM001	Ali	Hassan	Kolej Tuanku Canselor	Block S06	0123456789
UTM002	Abu	Bakar	Kolej Tun Hussein Onn	Block L07	0124598624
UTM004	Aisyah	Rahman	Kolej Perdana	Block D01	0192233445
UTM005	Daniel	Lim	Kolej 9	Block A1	0178899001
NULL	NULL	NULL	NULL	NULL	NULL

6.3 Test Query

6.3.1 View Customer Table

SELECT * FROM Customer;

UTMID	firstName	lastName	collegeName	block	customerPhoneNumber
UTM001	Ali	Hassan	Kolej Tuanku Canselor	Block S06	0123456789
UTM002	Abu	Bakar	Kolej Tun Hussein Onn	Block L07	0124598624
UTM004	Aisyah	Rahman	Kolej Perdana	Block D01	0192233445
UTM005	Daniel	Lim	Kolej 9	Block A1	0178899001
NULL	NULL	NULL	NULL	NULL	NULL

6.3.2 View Customer_Student Table

SELECT * FROM Customer_Student;

UTMID	StudentID	faculty	programme	yearOfStudy
UTM001	A22CS001	FKE	SKEE	2
UTM002	A22CS002	FC	SECR	3
UTM004	A23CS010	FC	SECJ	1
NULL	NULL	NULL	NULL	NULL

6.3.3 View Customer_UTMStaff Table

SELECT * FROM Customer_UTMStaff;

UTMID	UTMStaffID	position	department	officeLocation
UTM005	UTM8896	Lecturer	FC	N28-205
NULL	NULL	NULL	NULL	NULL

6.3.4 View Staff Table

SELECT * FROM Staff;

staffID	staffIC	staffFName	staffLName	staffEmail	staffLoginCredentials
STF001	950101015678	Siti	Aminah	siti@utm.my	hashed123
STF002	920505086543	Ahmad	Ali	ahmad@utm.my	hashed456
STF003	880707078888	Farah	Zainal	farah@utm.my	hashed789
NULL	NULL	NULL	NULL	NULL	NULL

6.3.5 View StaffTelephone Table

SELECT * FROM StaffTelephone;

	staffID	staffPhoneNumber
▶	STF001	0134567890
	STF002	0119876543
	STF003	0123344556
*	NULL	NULL

6.3.6 View Staff_CPPStaff Table

SELECT * FROM Staff_CPPStaff;

	staffID	staffRole	workShift
▶	STF001	CPP Staff	Morning
	STF003	CPP Staff	Evening
*	NULL	NULL	NULL

6.3.7 View Staff_Administrator Table

SELECT * FROM Staff_Administrator;

	staffID	adminLevel	accessPrivileges
▶	STF002	Level 1	Full Access
*	NULL	NULL	NULL

6.3.8 View SmartLocker Table

SELECT * FROM SmartLocker;

	lockerID	lockerStatus	capacity	locationName
▶	L001	Occupied	100	Kolej Tuanku Canselor
	L002	Available	80	Kolej Tun Hussein Onn
	L003	Occupied	60	Kolej Tun Fatimah
	L004	Available	50	Kolej Perdana
*	NULL	NULL	NULL	NULL

6.3.9 View LockerLocation Table

SELECT * FROM LockerLocation;

locationName
Kolej Dato Onn Jaafar
Kolej Rahman Putra
Kolej 10
Kolej 9
Kolej Perdana
Kolej Tuanku Canselor
Kolej Tun Dr. Ismail
Kolej Tun Fatimah
Kolej Tun Hussein Onn
NULL

6.3.10 View Parcel Table

SELECT * FROM Parcel;

parcelID	trackingNumber	courierName	parcelSize	arrivalDate	lockerID	staffID	parcelStatus	UTMID
P001	TRK123456	DHL	Small	2025-06-01	L001	STF001	Collected	UTM001
P002	TRK654321	J&T	Large	2025-06-02	L002	STF001	Collected	UTM002
P003	TRK888777	Shopee	Medium	2025-06-03	L003	STF003	Arrived	UTM004
P004	TRK999666	PosLaju	Small	2025-06-04	L004	STF003	Overdue	UTM005
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

6.3.11 View Notification Table

SELECT * FROM Notification;

notificationID	parcelID	UTMID	notificationDate	messageStatus
N001	P001	UTM001	2025-06-01	Sent
N002	P002	UTM002	2025-06-02	Sent
N003	P003	UTM004	2025-06-03	Sent
N004	P004	UTM005	2025-06-05	Failed
NULL	NULL	NULL	NULL	NULL

6.3.12 View Payment Table

SELECT * FROM Payment;

paymentID	parcelID	amount	paymentDate	staffID	NotificationID
PAY001	P002	5.00	2025-06-02	STF001	N002
PAY002	P004	3.00	2025-06-05	STF003	N004
NULL	NULL	NULL	NULL	NULL	NULL

6.3.13 View Report Table

SELECT * FROM Report;

reportID	reportDate	totalParcels	pendingParcels	collectedParcels	overdueParcels	totalRevenue	performanceSummary	staffID
R001	2025-06-30	120	10	100	10	250.00	Overall parcel collection is efficient	STF002
R002	2025-06-30	150	20	120	10	320.00	Parcel processing is efficient with minor delays	STF002
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

6.3.14 View ReportCategory Table

SELECT * FROM ReportCategory;

reportID	reportType
R001	Performance
R001	Revenue
NULL	NULL

6.3.15 View Report_Payment Table

SELECT * FROM Report_Payment;

reportID	paymentID
R001	PAY001
NULL	NULL

6.3.16 View Report_Parcel Table

SELECT * FROM Report_Parcel;

reportID	parcelID
R001	P001
R001	P002
NULL	NULL

6.3.17 View customer and their parcel

```
SELECT c.firstName, c.lastName, p.parcelID, p.parcelStatus
FROM Customer c
JOIN Parcel p ON c.UTMID = p.UTMID;
```

firstName	lastName	parcelID	parcelStatus
Ali	Hassan	P001	Collected
Abu	Bakar	P002	Collected
Aisyah	Rahman	P003	Arrived
Daniel	Lim	P004	Overdue

6.3.18 View available lockers

```
SELECT lockerID, locationName
FROM SmartLocker
WHERE lockerStatus = 'Available';
```

lockerID	locationName
L002	Kolej Tun Hussein Onn
L004	Kolej Perdana
NULL	NULL

6.3.19 Total revenue collected

```
SELECT SUM(amount) AS TotalRevenue
FROM Payment;
```

TotalRevenue
8.00

6.3.20 View all Smart Lockers with location

SELECT lockerID, lockerStatus, capacity, locationName
FROM SmartLocker;

lockerID	lockerStatus	capacity	locationName
L001	Occupied	100	Kolej Tuanku Canselor
L002	Available	80	Kolej Tun Hussein Onn
L003	Occupied	60	Kolej Tun Fatimah
L004	Available	50	Kolej Perdana
NULL	NULL	NULL	NULL

6.3.21 View parcels with customer name

```
SELECT
    p.parcelID,
    p.trackingNumber,
    CONCAT(c.firstName, ' ', c.lastName) AS customerName,
    p.parcelStatus
FROM Parcel p
JOIN Customer c ON p.UTMID = c.UTMID;
```

parcelID	trackingNumber	customerName	parcelStatus
P001	TRK123456	Ali Hassan	Collected
P002	TRK654321	Abu Bakar	Collected
P003	TRK888777	Aisyah Rahman	Arrived
P004	TRK999666	Daniel Lim	Overdue

6.3.22 Create View: Collected Parcels

```
CREATE VIEW View_CollectedParcels AS
SELECT parcelID, trackingNumber, arrivalDate
FROM Parcel
WHERE parcelStatus = 'Collected';
SELECT * FROM View_CollectedParcels;
```

parcelID	trackingNumber	arrivalDate
P001	TRK123456	2025-06-01
P002	TRK654321	2025-06-02

6.3.23 Display Customer Full Name Using CONCAT

```
CREATE VIEW View_CustomerParcels AS
SELECT
   c.UTMID,
   CONCAT(c.firstName, ' ', c.lastName) AS customerName,
   p.parcelID,
   p.parcelStatus
FROM Customer c
LEFT JOIN Parcel p ON c.UTMID = p.UTMID;
SELECT * FROM View_CustomerParcels;
```

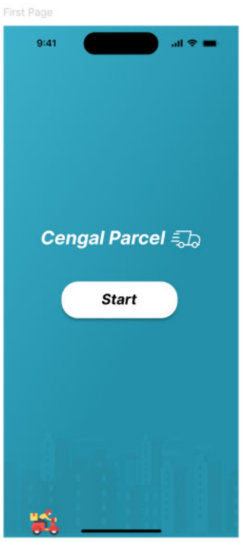
UTMID	customerName	parcelID	parcelStatus
UTM001	Ali Hassan	P001	Collected
UTM002	Abu Bakar	P002	Collected
UTM004	Aisyah Rahman	P003	Arrived
UTM005	Daniel Lim	P004	Overdue

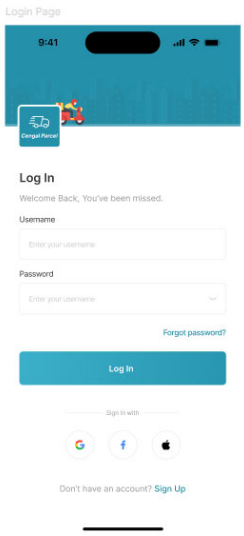
7.0 Interface Design

https://www.figma.com/design/dRdyZWcLJWr0RbIPjTb1TX/Courier---Tracking-App----App-UI--Community-?node-id=0-1&p=f&t=UJs2Bptqw6pHxFK4-0

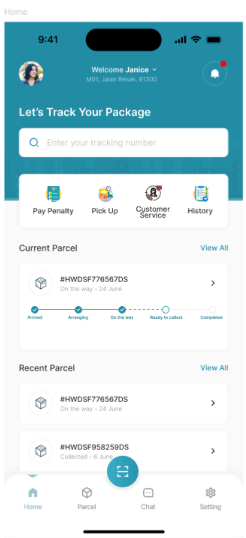
7.1 Customer Interfaces

7.1.1 Customer Login Page

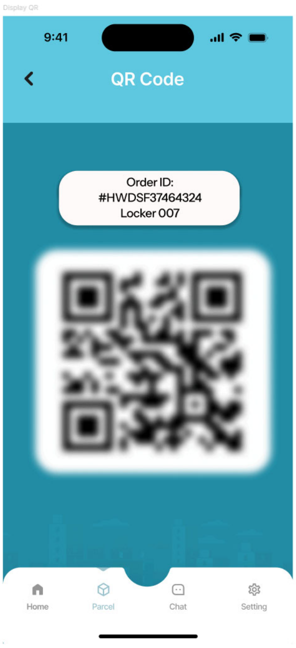


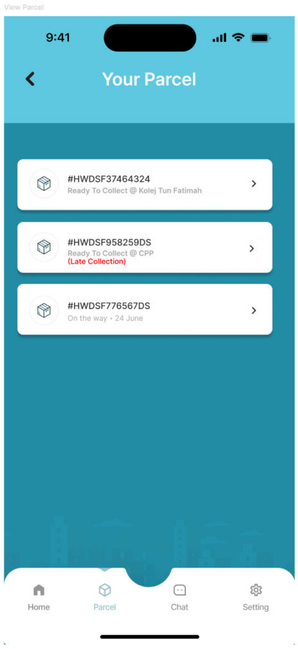


7.1.2 Customer Home Page

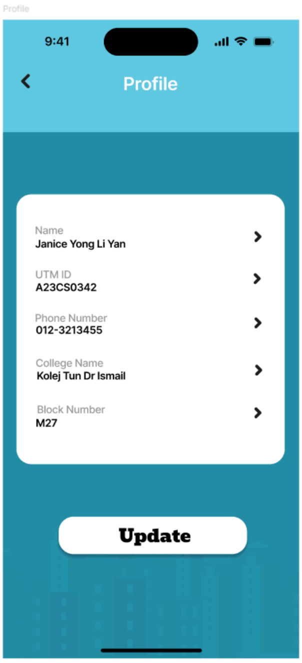


7.1.3 View Parcel

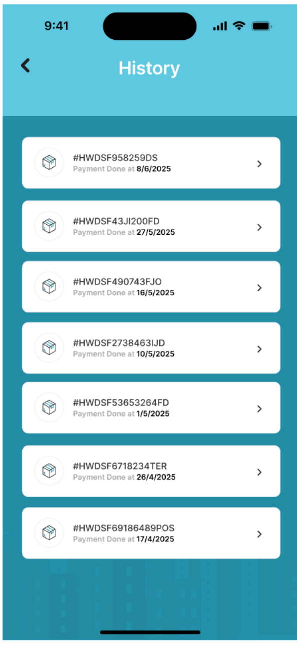




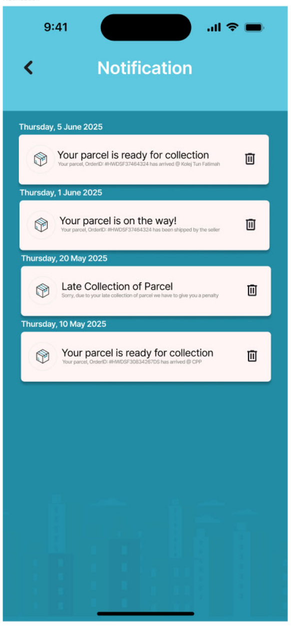
7.1.4 View Profile



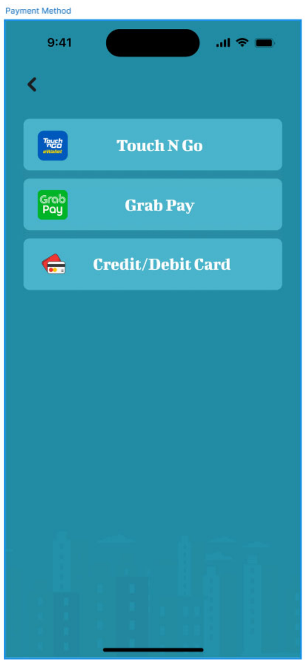
7.1.5 View History



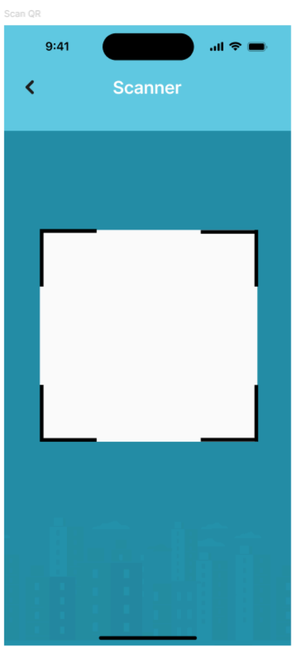
7.1.6 View Notification

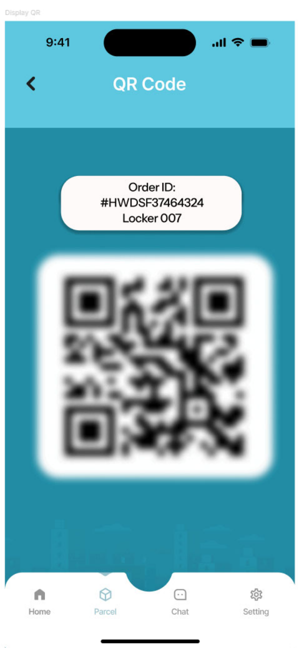


7.1.7 Make Payment



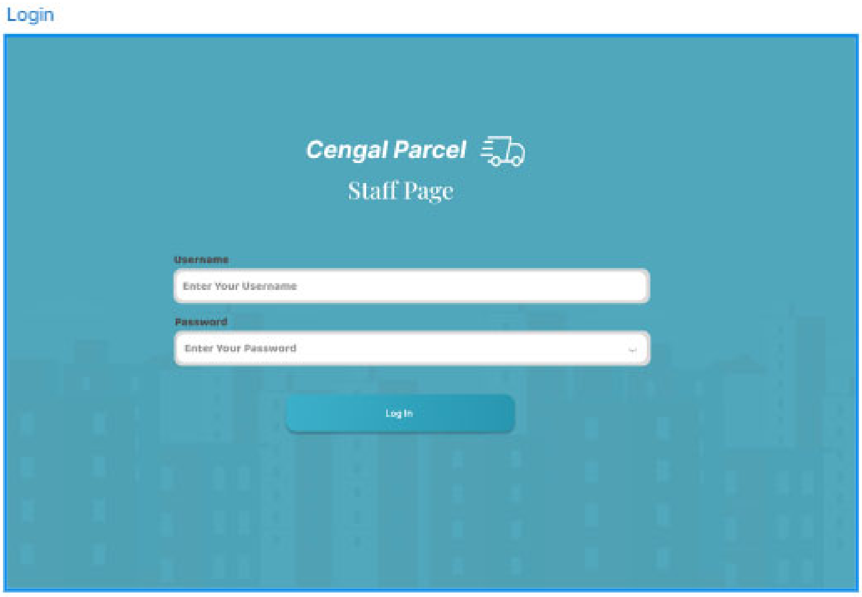
7.1.8 QR Code Scanning



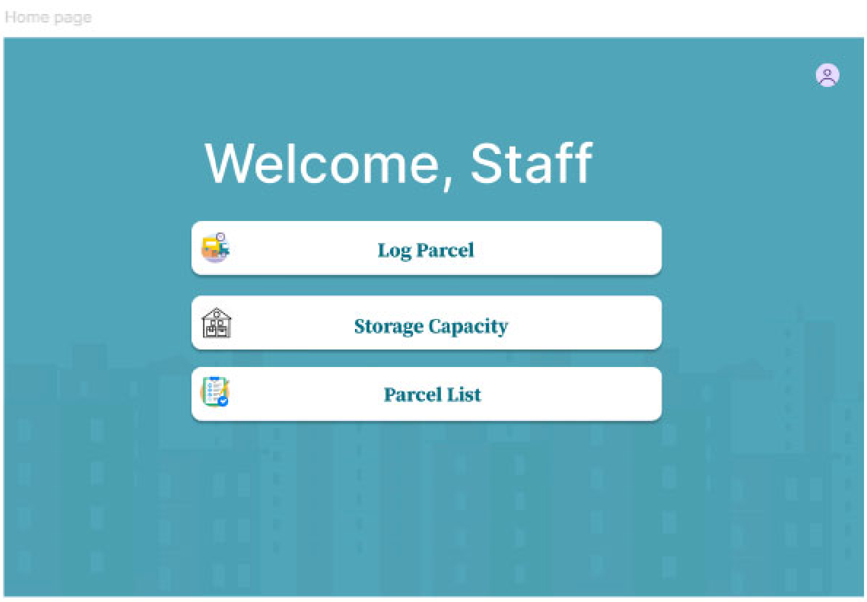


7.2 Staff Interfaces

7.2.1 Staff Login Page

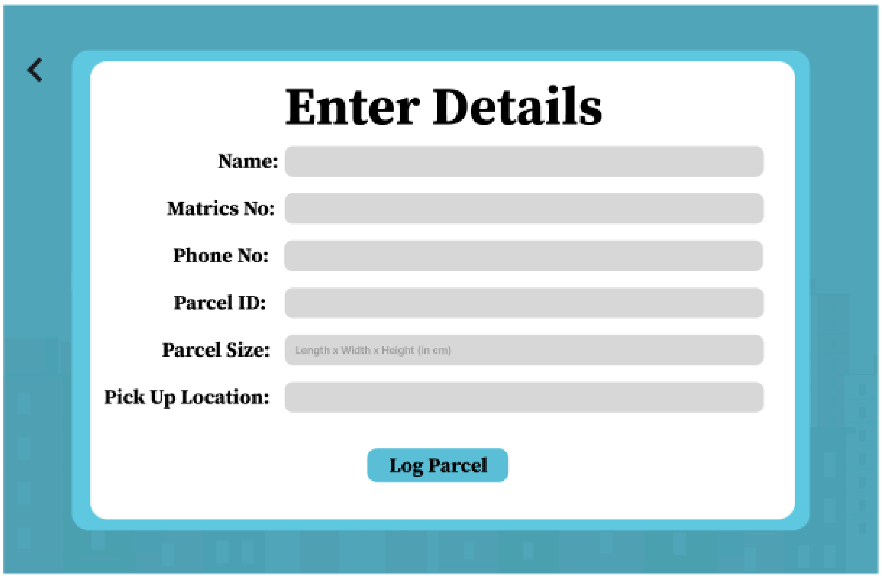


7.2.2 Staff Home Page



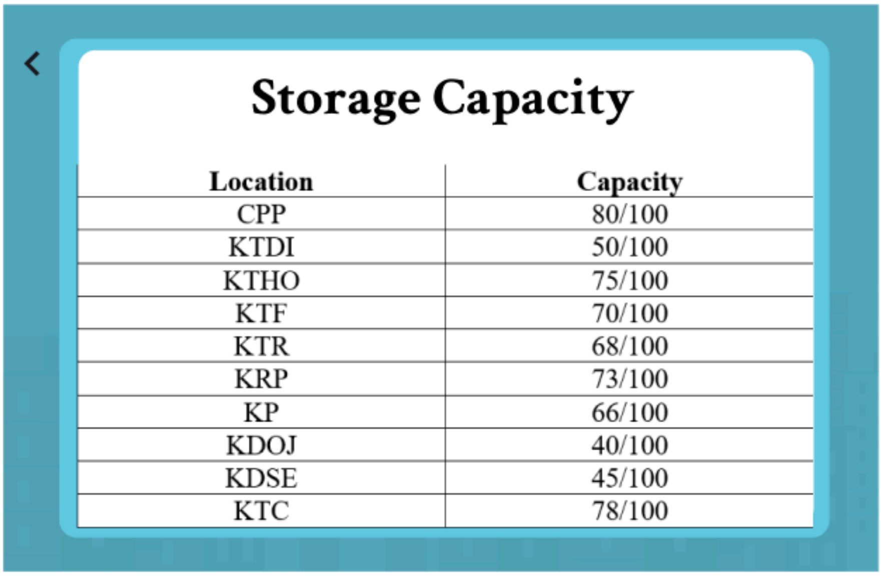
7.2.3 Log Parcel

Log Parcel



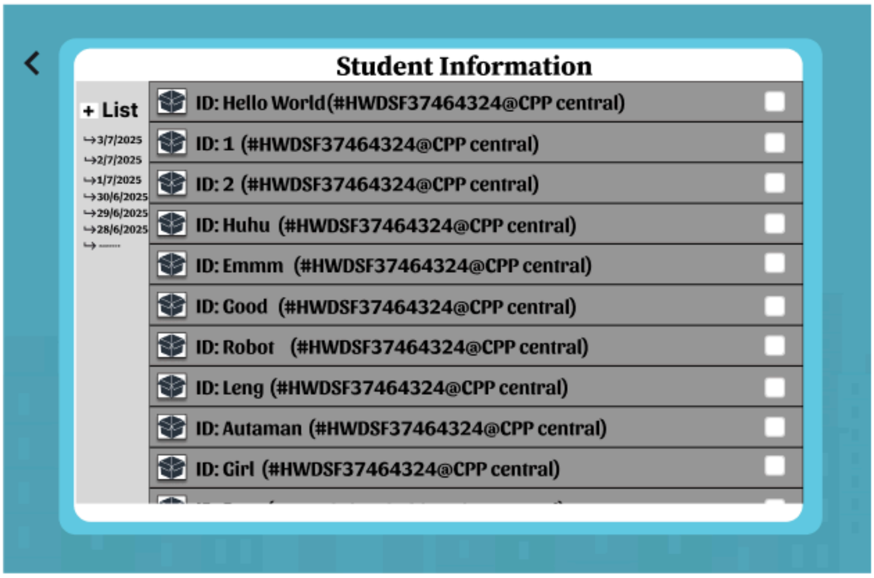
7.2.3 View Storage Availability

Capacity



7.2.4 View Parcel List

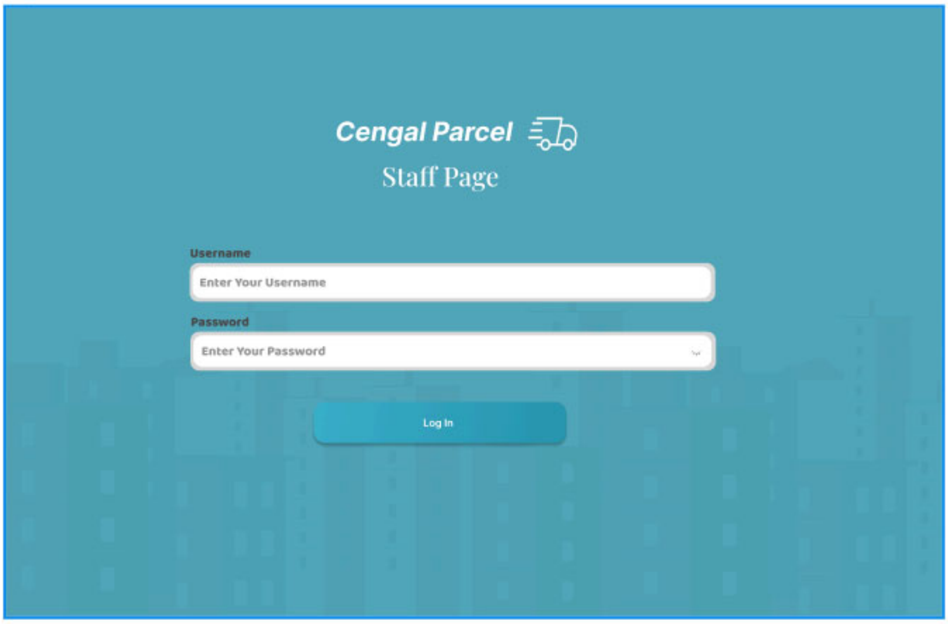
List



7.3 Administrative Interfaces

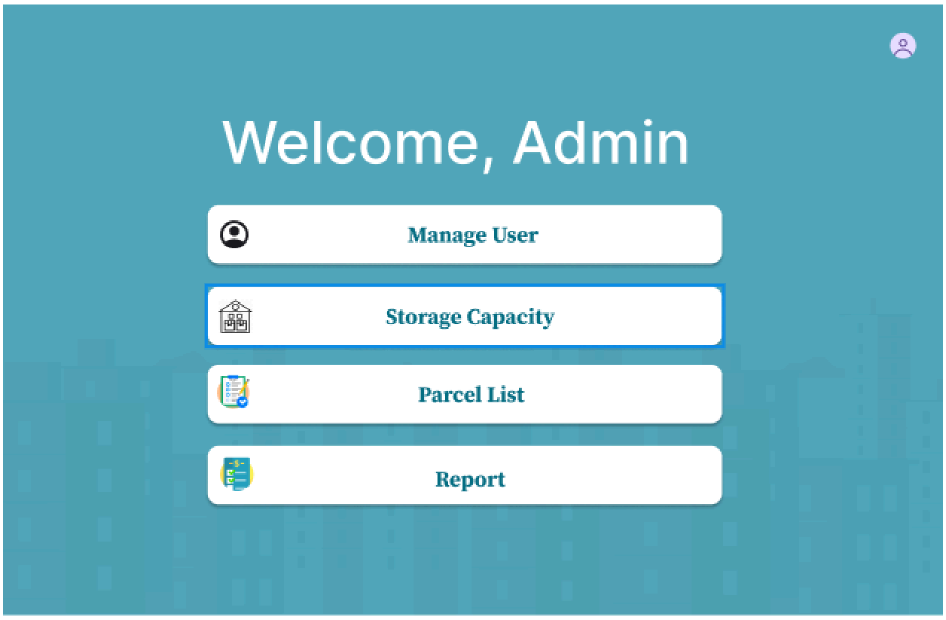
7.3.1 Admin Login Page

Login



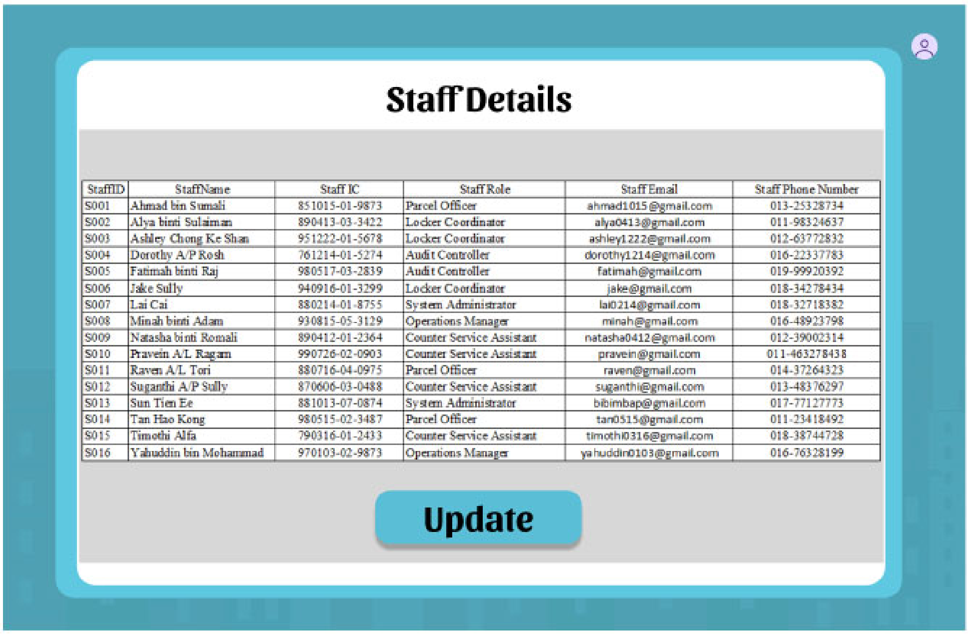
7.3.2 Admin Home Page

Home page



7.3.3 View and Update Staff Details

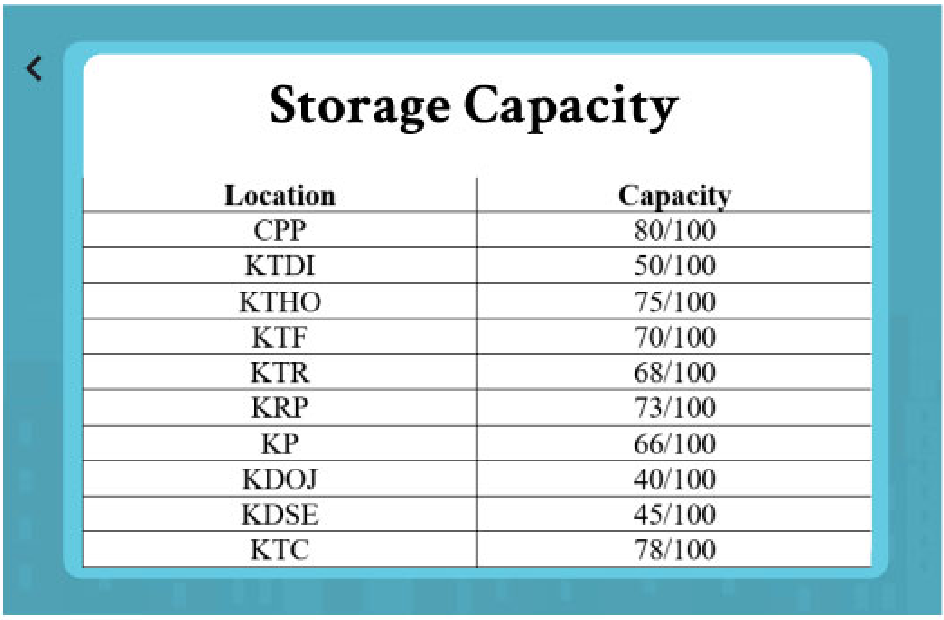
Manage User



StaffID	StaffName	Staff IC	Staff Role	Staff Email	Staff Phone Number
S001	Ahmad bin Sumali	851015-01-9873	Parcel Officer	ahmad1015@gmail.com	013-25328734
S002	Alya binti Sulaiman	890413-03-3422	Locker Coordinator	alya0413@gmail.com	011-98324637
S003	Ashley Chong Ke Shan	951222-01-5678	Locker Coordinator	ashley1222@gmail.com	012-63772832
S004	Dorothy A/P Rosh	761214-01-5274	Audit Controller	dorothy1214@gmail.com	016-22337783
S005	Fatimah binti Raj	980517-03-2839	Audit Controller	fatimah@gmail.com	019-99920392
S006	Jake Sully	940916-01-3299	Locker Coordinator	jake@gmail.com	018-34278434
S007	Lai Cai	880214-01-8755	System Administrator	lai0214@gmail.com	018-32718382
S008	Minah binti Adam	930815-05-3129	Operations Manager	minah@gmail.com	016-48923798
S009	Natasha binti Romali	890412-01-2364	Counter Service Assistant	natasha0412@gmail.com	012-39002314
S010	Pravein A/L Ragam	990726-02-0903	Counter Service Assistant	pravein@gmail.com	011-463278438
S011	Raven A/L Tori	880716-04-0975	Parcel Officer	raven@gmail.com	014-37264323
S012	Suganthi A/P Sully	870606-03-0488	Counter Service Assistant	suganthi@gmail.com	013-48376297
S013	Sun Tien Ee	881013-07-0874	System Administrator	bibimbap@gmail.com	017-77127773
S014	Tan Hao Kong	980515-02-3487	Parcel Officer	tan0515@gmail.com	011-23418492
S015	Timothi Alfa	790316-01-2433	Counter Service Assistant	timothi0316@gmail.com	018-38744728
S016	Yahuddin bin Mohammad	970103-02-9873	Operations Manager	yahuddin0103@gmail.com	016-76328199

7.3.4 View Storage Availability

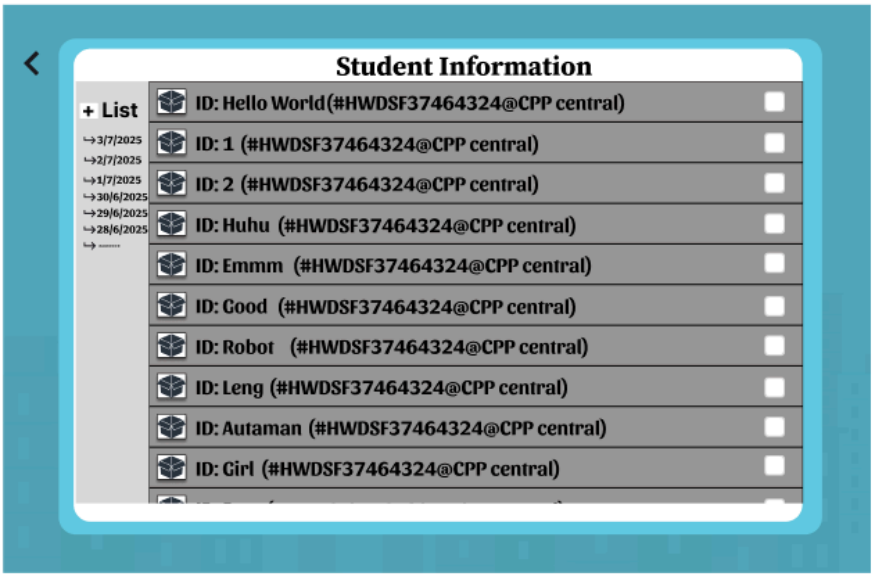
Capacity



Location	Capacity
CPP	80/100
KTDI	50/100
KTHO	75/100
KTF	70/100
KTR	68/100
KRP	73/100
KP	66/100
KDOJ	40/100
KDSE	45/100
KTC	78/100

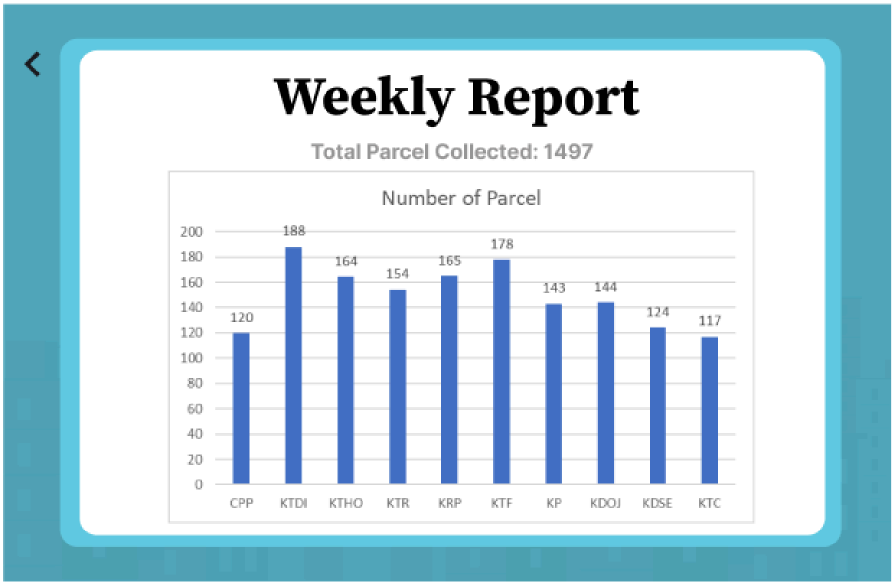
7.3.5 View Parcel List

List



7.3.6 View Report

Report



8.0 Summary

The Parcel Service System project was initiated to address the critical logistical inefficiencies faced by the University Center for Parcel Processing (CPP). By transitioning from a manual, fragmented recording system to a centralized, automated database solution, the project aims to significantly enhance the speed, accuracy, and security of parcel management on campus.

Throughout the development of this project, a rigorous database design methodology was applied. We began by constructing a Logical Entity-Relationship Model that accurately represented complex real-world relationships, including the implementation of Superclass/Subclass hierarchies for Users and Management staff to handle role-specific attributes efficiently. To ensure data integrity and eliminate redundancy, the schema underwent a strict Normalization process, refining all 14 entities from First Normal Form (1NF) up to Boyce-Codd Normal Form (BCNF). This process successfully resolved issues related to composite attributes (splitting names and addresses) and multi-valued attributes (creating separate relations for staff contact numbers and report types).

The final outcome of this phase is a robust, fully normalized Relational Database Schema ready for physical implementation using SQL. The system is designed to support high-volume transactions, including automated smart locker assignments, real-time customer notifications, and secure financial processing. By enforcing strict business rules, such as valid location constraints for lockers and distinct role-based access controls, the proposed system not only solves the immediate problem of parcel congestion but also provides a scalable foundation for future operational growth within the university.